BML for Communicating with Multi-Robot Systems

Conference Paper · September 2015

4 authors:

Thomas Remmersmann
Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE

Ulrich Schade
Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE

Kellyn Rein
Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE

Alexander Tiderko
Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE

BML for Communicating with Multi-Robot Systems

Thomas Remmersmann, Dr. Ulrich Schade, Kellyn Rein, and Alexander Tiderko

Fraunhofer FKIE
Fraunhofer Straße 20
53343 Wachtberg, Germany
+49 228 9435376
thomas.remmersmann@fkie.fraunhofer.de

Keywords
C-BML, Interoperability, Command and Control, Multi Robot Systems

ABSTRACT: *When BML was conceived for C2-to-simulation interoperation back in 2000, the idea to control robotic autonomous forces using BML had already been considered. Over the past few years, starting in 2010, we have at last implemented this idea and we are running several projects focusing on multi-robot systems controlled by BML.*

In this paper, we present the concepts of how to implement command of multi-robot systems using BML. In addition, we discuss how BML, developed for C2-to-simulation interoperation needed to be adjusted to fit the requirements for commanding multi-robot systems. In order to command multi-robot systems we created a control/planning node that receives high-level tasks in BML and disaggregates them into simple BML tasks such to move or to take a picture of an object being observed (image intelligence gathering). This node also takes care of the reporting. For example, it may receive task status reports for several ongoing basic tasks and from this it can calculate the aggregated task status report for the corresponding high-level task. Along with task status reports, general status reports and position reports have been dealt with. However, there are still many more aspects which robots can report on that might be relevant for users. Since we are using reconfigurable robots, the user may need to know the current configuration of each robot which thus is to be conveyed by a report. For these kinds of reports we use the so-called "WhoHolding" reports. Of course, the robots also have to report sensor data collected. Therefore, we have developed reports for pictures taken, videos recorded, and sensor measurements such as temperature, gas concentrations etc. For all these kinds of information we have extended the BML schemata taking the general BML "look and feel" into account.

1. Introduction

Commanding and controlling a multi-robot system (MRS) is a challenging task with the complexity increasing progressively as the number and the heterogeneity of robots in the system increases. Our approach is to handle the complexity of an MRS operation by using a BML based interactive command and control (C2) system. It gives a human operator the ability to express the commands on an abstract level and therefore is able to focus on mission-critical tasks, while detailed planning and execution is directed to a computer system. To prevent an information overflow from the mass of reports that all the robots with all their sensors may generate, we implemented an aggregation process to create high level reports to send the user only the information she needs.

The rest of the paper is structured as follows. In Section 2 we present an overview over the development of BML in general and on work concerning BML for commanding multi-robot systems in particular. In Section 3 we describe and discuss the structure of the communication with the robots we used in our approach. Sections 4 and 5 are about the specifics of communicating with multi-robot systems using BML. In particular, Section 4 is about commanding robots via BML, and Section 5 is about reporting. In Section 6 we briefly state our conclusions and outline possible future work.

2. Battle Management Language (BML)

In this section, we will first provide an overview over BML in general (Subsection 2.1). After that, we will present a summary of some of our previous papers and one other paper on commanding multi-robots systems with BML (Subsection 2.2) in order to prepare the discussion in the following sections.

2.1. Battle Management Language for the Interaction between C2 Systems and Simulation Systems

Primary, BML serves as a tool to automate and facilitate the use of simulation systems in staff training, after-action analysis, and decision support to name a few uses. The concept was first envisioned around the year 2000 [1] and began in work sponsored by the US Army's Simulation-to-C4I Interoperability Overarching Integrated Product Team (SIMCI OIPT) [2]. As a result, SISO appointed a BML Study Group that produced a report in 2005 [3], opening the road for the Product Development Group (PDG) "Coalition BML". At the same time, NATO RTO Modeling and Simulation Group recognized the need for a formal language to exchange orders and reports between C2 systems and simulation systems and thus created MSG-048 "Coalition Battle Management Language" to explore the promise of BML. MSG-048, which was awarded the NATO STO 2013 Scientific Achievement Award, ran from 2006 to 2010. After waiting some length for proposals for a BML standard from SISO's PDG, MSG-048 developed their own versions of BML and successfully tested those in a series of experiments. The results of those experiments are summarized in [4]. The work of MSG-048 was continued by MSG-085 "Standardization for C2-Simulation Interoperation". Finally, the SISO PDG approved a BML Phase 1 Full Schema, which we used for our experiments and which was the starting point for the extensions we present in this paper.

2.2. Battle Management Language for Communicating with Robots

As previously mentioned, the idea of controlling robotic forces with BML has been around since 2000. But there was not much research on this topic up to 2010. We started the research on this topic in 2010 and presented a system where real robots executed BML commands [5]. Similar research has been done on controlling simulated Unmanned Air Vehicles (UAVs) by Heffner and Hassaine [6]. Langerwisch et al. described a larger BML-based robot control experiment in 2012 [7]. On the basis of these experiences and their results, we developed a disaggregation and planning system for BML tasks and described it in the papers [8][9].

3. System Structure

In this section, we will describe and discuss the general structure behind our approach. BML is essential for this approach as it allows expressing orders and reports on an abstract level. Orders expressed on this level cannot be processed directly by robots, and robots cannot send reports on that level without additional supporting software. However, the abstract level of BML allows a single operator to command the whole multi-robot system (the MRS) since the operator is freed from the burden of handling all the details of the robotic actions such as collision avoidance.

So, how does our approach work? BML orders are sent to the MRS control. The MRS control has its own intelligence that determines the details about how to execute the orders taking the different capabilities of the robots into account. This corresponds to the interpretation of BML orders by a simulation system. In short, the MRS control uses its own intelligence for disaggregation of the given order, for planning how the goals determined by that order can be achieved, and for sending the

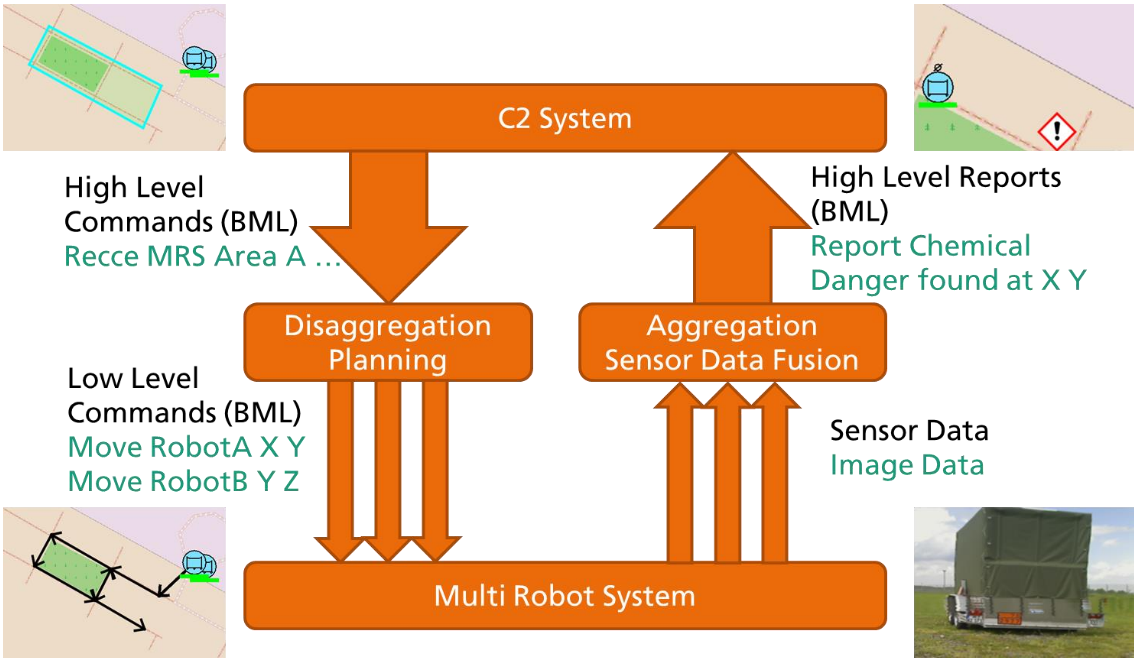

Figure 1: System architecture, disaggregation of high-level commands to low-level commands on the left side and aggregation of information from sensor data to high-level reports on the right.

disaggregated commands to the robots in the MRS. Details of the planning and disaggregation process are given in Section 4.

The other way around, BML reports are sent form the MRS to the controller's GUI. Like the orders, the reports also include a degree of abstraction in order to provide the controller with exactly the information she should be aware of, suppressing irrelevant details. This again is analogous to the communication with simulation systems for which, for example, the rate of positional reports from the simulated units needed to be dropped significantly [10]. So, the MRS is outfitted with enough intelligence to take data provided by the MRS's sensors (regardless of which specific robot and which specific sensor attached to that robot originally provided the data) and aggregate these data into information that can be expressed as BML reports which can then be sent to the C2 system. Details on that aggregation process are given in Section 5.

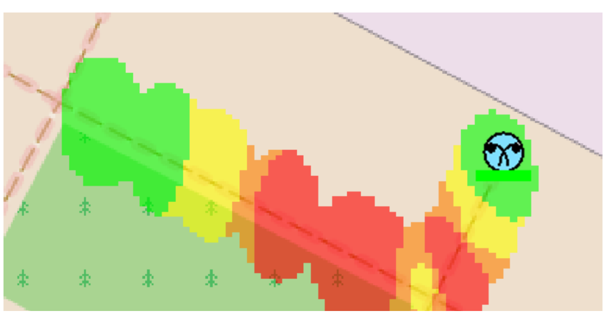

Figure 2: Interpolation of gas concentration reported as a Resource Report with a spatial reference.

A graphic representation of the system with the disaggregation of orders and aggregation of reports along with examples can be seen in Figure 1.

4. Commanding / Disaggregation

In order to allow a single operator to command the whole MRS, we gave him the possibility to give commands on an abstract level. We then developed a control node that takes care of the disaggregation of the command and of the execution control. When, for example, a commander needs to recce the streets in a certain area he creates a "recce" task for the MRS specifying the area of interest. The control node then breaks that command into low-level commands, in this case, into simple move commands, e.g., one move command for each street in that area. This example is visualized on the left side of Figure 1. Next, a planning system decides which robot should execute

```
<cbml:Report xsi:type="cbml:ResourceType">
	<cbml:ReporterWho>
		<cbml:OrganisationRef xsi:type="cbml:UnitRef">
			<cbml:OID>Robot_A</cbml:OID>
		</cbml:OrganisationRef>
	</cbml:ReporterWho>
	<cbml:ReportedWhen xsi:type="cbml:ReportedWhenAbsoluteTimingType">
		<cbml:ReportingDatetime>20121212134533.221</cbml:ReportingDatetime>
	</cbml:ReportedWhen>
	<cbml:ReportingData>
		<cbml:OID>rep_1234</cbml:OID>
		<cbml:ReportingDataCategoryCode>REP</cbml:ReportingDataCategoryCode>
	</cbml:ReportingData>
	<cbml:ObjectTypeRef xsi:type="cbml:GeographicFeatureTypeRef">
		<cbml:OID>point_1234</cbml:OID>
	</cbml:ObjectTypeRef>
	<cbml:ResourceURL>http://ourserver/url/to/pic.jpg</cbml:ResourceURL>
</cbml:Report>
```

Listing 1: Example of ResourceType Report. Robot_A reports that it has taken a picture at a certain point and that is available under the given URL.

which move command, taking into account the goal of executing all commands as quickly as possible. Finally, the selected robots are tasked to move to the road to which they are currently assigned, in order to start the recce process. During execution, the control node continuously checks whether the assignment of robots to specific tasks is still efficient. If, say, a robot is slowed down by obstacles, the assignment might be changed.

Disaggregation of high-level commands is a complex task, depending upon a lot of factors such as the number of robots available, the equipment and capabilities of those robots and specifics of the scenario. An approach to describing the disaggregation with an extended BML dialect is provided in [9].

5. Reporting / Aggregation

During execution of the aggregation process, the control node works like an officer. It receives the reports originating from low-level tasks that were created during the disaggregation process and from those it generates reports about the high-level task it originally received from the controller. In particular, this means if all low-level tasks belonging to one high-level task are completed, a report about the completion of the high level task is generated.

It turned out that we needed more kinds of reports to fit our needs. In particular, reports are needed by which new binary data, e. g. images or videos, are made available to the controller. Since images or videos themselves cannot be expressed directly using human-readable text, but must first be processed or analyzed by a human or an algorithm, we cannot provide information on their content directly in a BML message. Instead, we created a new kind of report called “ResourceType”. In this report type we reference the URL from which the binary data (the image or the video stream) can be requested. We also added a geographic reference to this new report type, so we can add the location at which the image was taken or what area the image shows. An example report is shown in Listing 1.

Another report which we introduced [11] is the “WhoMeasuredType” report, which allows report about sensor measurements of any kind. It includes the measurement by value, the unit of measurement (UOM) and indicates the phenomenon which was measured. Such a phenomenon can be, for example, air temperature or gas concentration as well as the internal battery status of the robot. The report also includes where and when the measurement was taken, as well as naming the sensor that was used for that measurement. An example of this can be

```xml
<cbml:Report xsi:type="cbml:WhoMeasuredType">
	<cbml:ReporterWho>
		<cbml:OrganisationRef xsi:type="cbml:UnitRef">
			<cbml:OID>Robot_A</cbml:OID>
		</cbml:OrganisationRef>
	</cbml:ReporterWho>
	<cbml:ReportedWhen xsi:type="cbml:ReportedWhenAbsoluteTimingType">
		<cbml:ReportingDatetime>20121212134533.221</cbml:ReportingDatetime>
	</cbml:ReportedWhen>
	<cbml:ReportingData>
		<cbml:OID>123</cbml:OID>
		<cbml:ReportingDataCategoryCode>REP</cbml:ReportingDataCategoryCode>
	</cbml:ReportingData>
	<cbml:WhoRef>
		<cbml:ObjectItemRef xsi:type="cbml:OtherObjectItemRef">
			<cbml:OID>TemperatureSensor12</cbml:OID>
		</cbml:ObjectItemRef>
	</cbml:WhoRef>
	<cbml:ObjectTypeRef xsi:type="cbml:GeographicFeatureTypeRef">
		<cbml:OID>point_1232453</cbml:OID>
	</cbml:ObjectTypeRef>
	<cbml:Measurement>
		<cbml:Phenomenon>temperature</cbml:Phenomenon>
		<cbml:Value>25</cbml:Value>
		<cbml:UOM>°C</cbml:UOM>
		<cbml:TimeOfMeasurement>20131212134522.123</cbml:TimeOfMeasurement>
	</cbml:Measurement>
</cbml:Report>
```


Listing 2: Example of WhoMeasuredType Report, reporting that Robot_A has measured a temperature of 25°C at a certain point at the given time with a sensor called "TemperatureSensor12".

seen in Listing 2.

Information about measurements can be aggregated. For example, in the case of gas concentrations we did a spatial interpolation over the single measurements and reported this interpolation, refer to Figure 2 for the resulting image of such an interpolation. The aggregation process receives many measurement reports, then processes the reports and uses the previously mentioned "ResourceType" report with a bounding box as a geographic reference to report the interpolated data.

As mentioned in Section 4, for efficient and effective planning it is necessary to know about the equipment and capabilities of the robots [11]. Since we were using reconfigurable robots, we solved this problem by letting the robots report about all the equipment that is mounted on them. This was done using the "WhoHoldingType". The reports from a MRS provide the planning system with the necessary information to deduce which robot is able to execute which task. An example report about a 3DScanner is shown in Listing 3.

6. Conclusion and Outlook

We have shown how we use BML to control multi-robot systems and how BML can be used in the aggregation and disaggregation processes. We also introduced two new types of reports – the measurement report and the resource report – and discussed some use cases for them. These reports might also be used in C2-to-simulation scenarios, e.g., the simulation might send reports with computer generated images as a result of a simulated reconnaissance task which would increase realism in a simulation supported staff training. Of course, measurement reports

```xml
<cbml:Report xsi:type="cbml:WhoHoldingType">
	<cbml:ReporterWho>
		<cbml:OrganisationRef xsi:type="cbml:UnitRef">
			<cbml:OID>Robot_A</cbml:OID>
		</cbml:OrganisationRef>
	</cbml:ReporterWho>
	<cbml:ReportedWhen xsi:type="cbml:ReportedWhenAbsoluteTimingType">
		<cbml:ReportingDatetime>20132412185322.111</cbml:ReportingDatetime>
	</cbml:ReportedWhen>
	<cbml:ReportingData>
		<cbml:OID>1</cbml:OID>
		<cbml:ReportingDataCategoryCode>REP</cbml:ReportingDataCategoryCode>
	</cbml:ReportingData>
	<cbml:WhoRef>
		<cbml:ObjectItemRef xsi:type="cbml:UnitRef">
			<cbml:OID>Robot_A</cbml:OID>
		</cbml:ObjectItemRef>
	</cbml:WhoRef>
	<cbml:ObjectTypeRef xsi:type="cbml:ElectronicEquipmentTypeRef">
		<cbml:OID>123456789</cbml:OID>
	</cbml:ObjectTypeRef>
	<cbml:Holding>
		<cbml:OID>urn:fkie:fraunhofer:sensor:3DScanner</cbml:OID>
		<cbml:OperationalCount>1</cbml:OperationalCount>
	</cbml:Holding>
</cbml:Report>
```

Listing 3: Example of WhoHoldingType Report. Robot_A reports that it has one 3D scanner.

can be used to report any kind of measurement that might be relevant in the simulation.

7. References

[1] J. Abbott, S. Levine, and M. Pullen, “Answering the Question: Why a BML Standard has Taken So Long to be Established?” 2011 Fall Simulation Interoperability Workshop (= paper 11F-SIW-020), Orlando, FL, 2011.

[2] S. Carey, M. Kleiner, M.R. Hieb, and R. Brown, “Standardizing Battle Management Language – A Vital Move towards the Army Transformation” Fall Simulation Interoperability Workshop (= paper 01F-SIW-067), Orlando, FL, 2001.

[3] C. Blais, K. Galvin, and M.R. Hieb “Coalition Battle Management Language (C-BML) Study Group Report” IEEE Fall Simulation Interoperability Workshop, Orlando FL, 2005.

[4] K. Heffner, A. Brook, N. de Reus, L. Khimeche, O.M. Mevassvik, M. Pullen, U. Schade, J. Simonsen, and R. Gomez-Veiga, “NATO MSG-048 C-BML Final Report Summary” 2010 Fall Simulation Interoperability Workshop (= paper 10F-SIW-039), Orlando, FL, 2010.

[5] T. Remmersmann, B. Brüggemann, and M. Frey “Robots to the ground.” In Concepts and Implementations for Innovative Military Communications and Information Technologies, pages 61–68. Military University of Technology, September 2010.

[6] K. Heffner and F. Hassaine, “Towards intelligent operator interfaces in support of autonomous uvs operations," in 16th ICCRTS - Collective C2 in Multinational Civil-Military Operations, 2011.

[7] M. Langerwisch, M. Ax, S. Thamke, T. Remmersmann, A. Tiderko, and B. Wagner “Realization of an Autonomous Team of

Unmanned Ground and Aerial Vehicles”, 2012 International Conference on Intelligent Robotics and Applications (ICIRA 2012), Montréal, Canada, October 2012.

[8] T. Remmersmann, U. Schade, and C. Schlick, Using Quasi-Natural Language Sentences for Supervisory Control of Multi-Robot Systems, 2012 IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2012), Seoul, Korea, October 2012.

[9] T. Remmersmann, U. Schade, and C. Schlick “Interactive Multi-Robot Command and Control with Quasi-Natural Command Language” IEEE International Conference on Systems, Man, and Cybernetics, San Diego, CA, October 2014.

[10] R. El Abdouni Khayari, L. Khimeche, H. B. Lotz, X. Cuneo, U. Krosta, T. Remmersmann “Practical Use of BML and MSDL Standards for Supporting French German Training”, 2015 Fall Simulation Interoperability Workshop (= Paper 15F-SIW-014), Orlando, FL, 2015.

[11] T. Remmersmann, U. Schade and A. Tiderko “Commanding Heterogeneous Multi-Robot Teams”, 19th International Command and Control Research and Technology Symposium (ICCRTS), Alexandria, VA, 2014.

8. Author Biographies

THOMAS REMMERSMANN is a research associate and Software Engineer with Fraunhofer FKIE in Wachtberg, Germany. He has several years of experience with building and evaluating C2 interfaces to interact with simulation systems and robotic forces, especially with the use of BML.

Dr. ULRICH SCHADE is a senior research scientist with Fraunhofer FKIE and is associate professor to the Institute for Communication Science, Bonn University. He is an expert in computational linguistics and has contributed greatly to the understanding of how formal grammar can improve the development of BML.

KELLYN REIN is a research associate with Fraunhofer FKIE. Her main research focus is on multi-level multisource information fusion using BML as a basis, with a particular emphasis on the evaluation of uncertainty.

ALEXANDER TIDERKO obtained his Dipl.-Inform. degree in computer science from the University of Bonn, Germany in 2006. At present he is a scientist at the Fraunhofer Institute for Communication, Information Processing and Ergonomics FKIE in Wachtberg Germany. His primary research interests include networked multi-robot systems and robot interoperability.